

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CE-6002-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Structural Design - I (RCC)

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume missing data, if any suitability.

1. a) Discuss various assumptions of limit state design method. 7
b) Discuss balanced section with regard to working stress design method. 7
2. Design a rectangular beam for an effective span of 5m. The superimposed load is 60kN/m and size of the beam is limited to 23cm×45cm overall. Use M₂₀ concrete and Fe 415 grade steel. Consider the exposure condition as moderate. 14
3. Design a slab for a room 8.0m × 4.0m clear in size if the live load is 2kN/m². The slab is simply supported on all four edges. Exposure condition is mild. 14
4. Design a short column subjected to axial load of 1000 kN. Moment in one direction as 60 kN-m and moment in other direction as 40kN-m. 14

CE-6002-CBGS

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[2]

5. Design a dog-legged staircase to be provided in a residential multi-storeyed building. Clear space available is $2.4\text{m} \times 4.8\text{m}$. Height of each flight is 1.5m and floor to floor height is 3m . Length of landing on either side along the direction of flight is 1m . Take live load as 3.0kN/m^2 . 14
6. Design a combined rectangular footing for two columns 3m apart. Sizes of the columns are $300\text{mm} \times 300\text{mm}$ and $400\text{mm} \times 400\text{mm}$ are carrying loads 300kN and 600kN respectively. The safe bearing capacity of soil is 150kN/m^2 . 14
7. Determine the ultimate moment capacity of the following T beam : Effective flange width is 900mm . Depth of flange is 140mm , web width is 300mm , effective depth is 420mm and area of tension steel is 1570mm^2 . Assume Fe 415 steel and M25 grade concrete. 14
8. Write short notes on the following: 14
 - i) Circular slabs
 - ii) Waffle slabs
 - iii) Flat slabs
 - iv) Yield line theory
